



N O R M A T I V E

$T = I \quad \det = +1 \quad (x, y) \rightarrow (x, y)$

"This is the standard."

RFC 2119 MUST / SHALL

Owl Semaphore

System Specification — A Finite Algebra of Epistemic States

Carey James Balboa

Independent DNS Security Researcher

ORCID 0009-0000-5237-9065 CONCEPT-DOI 10.5281/zenodo.19473697
PUBLISHED-VERSION-DOI 10.5281/zenodo.19474599 (v1.2.0) RC-VERSION-
DOI TBD_BY_ZENODO_ON_RELEASE
SOURCE github.com/IT-Help-San-Diego/owl-semaphore VERSION v1.3.0-rc · LICENSE CC-BY-4.0

BANNER-TUPLE :: STATE=SYSTEM :: TRANSFORM=T = I $\det = +1$ $(x, y) \rightarrow (x, y)$:: QUOTE="This is the standard." :: VERSION=v1.3.0-rc :: CONCEPT-DOI=10.5281/zenodo.19473697 :: PUBLISHED-VERSION-DOI=10.5281/zenodo.19474599 :: RC-VERSION-DOI=TBD_BY_ZENODO_ON_RELEASE

OWL SEMAPHORE SYSTEM — MASTER PROOF



A finite algebra over epistemic states, implemented as a reproducible visual notation system with enforced invariants.

1. Statement of Intent

This document defines the Owl Semaphore as a formal epistemic system grounded in mathematics, perception, and reproducible graphical structure.



This is not a taxonomy of labels. It is a closed algebra over epistemic states, mapped into a constrained visual notation system.

The objective is a system that is:

- mathematically coherent
- visually invariant
- epistemically meaningful
- operationally reproducible
- resistant to ambiguity and drift

1.1 Canonical Sentence Stack (v1.3.0-rc)

The project uses a three-layer canonical sentence stack so that the same concept can be expressed at the level of mathematics, operation, and human story without drifting between documents:

Layer	Canonical sentence	Use
Formal	<i>A finite algebra over epistemic states, implemented as a reproducible visual notation system with enforced invariants.</i>	README, system spec, Zenodo meta-data draft, citation abstract
Operational	<i>A four-state visual system for marking how a claim, document, dataset, or finding should be evaluated before belief, challenge, or action.</i>	Explanation document, public overview, DNS Tool bridge
Human	<i>Four owls tell the reader what kind of thinking they are looking at: standard, exploration, inversion, or self-audit.</i>	Story sections, teaching material

Earlier inconsistent forms — “mapped into a visual system with strict invariants” (former §11) and “implemented as a reproducible visual system with enforced invariants” (former README masthead) — are reconciled in v1.3.0-rc onto the canonical formal sentence above.

2. Mathematical Foundation

2.1 Group Definition

The Owl Semaphore’s discrete state space is modeled as the Klein four-group:

$$V_4 = \{I, \sigma_v, C_2, \sigma_h\}$$



This is a finite subgroup of the orthogonal group $O(2)$ isomorphic to V_4 (equivalently the dihedral group D_2); it is not $O(2)$ itself (Klein four-group, Wikipedia; nLab; Knill, Harvard Math 22b, Unit 8: $O(2)$).

2.2 Elements

State	Operator	Mapping	Determinant
NORMATIVE	I	$(x,y) \rightarrow (x,y)$	$+1$
NON-NORMATIVE	σ_v	$(x,y) \rightarrow (-x,y)$	-1
CRITICAL	C_2	$(x,y) \rightarrow (-x,-y)$	$+1$
METACOGNITIVE	σ_h	$(x,y) \rightarrow (x,-y)$	-1

The σ_h assignment to METACOGNITIVE is unchanged from v1.2.0. Only the interpretive wording is refined in v1.3.0-rc (see §4).

2.3 Closure (Cayley Table)

The system is closed under composition. The Cayley table is:

$\circ$	I	σ_v	C_2	σ_h
I	I	σ_v	C_2	σ_h
σ_v	σ_v	I	σ_h	C_2
C_2	C_2	σ_h	I	σ_v
σ_h	σ_h	C_2	σ_v	I

Each element is its own inverse: $g^2 = I$ for all $g \in V_4$. The four group axioms (closure, associativity, identity, inverses) hold by the table above.

2.4 Interpretation

This closure is not decorative. It enforces that all epistemic transitions remain inside a defined state space. Group structure guarantees algebraic closure; it does not by itself guarantee security or behavioral correctness without further proof.

3. State vs Process

3.1 Discrete States

The four owls represent discrete epistemic states:

- identity (NORMATIVE)



- reflection (NON-NORMATIVE)
- inversion (CRITICAL)
- frame-audit (METACOGNITIVE)

3.2 Continuous Process

Not all operations belong to the state system.

3.3 The 31° Rotation

The measured $\sim 31^\circ$ rotation is not part of V_4 .

- it is not closed
- repeated composition does not return to the set

It represents **process**, not state.

3.4 Principle

States classify position. Processes move between positions.

4. Epistemic Model

4.1 Core Structure

The system separates three levels:

1. object of analysis
2. observer
3. evaluative frame

4.2 State Mapping (Normative Phrasing — v1.3.0-rc)

State	Quote (scientific / normative)	Meaning
NORMATIVE	<i>“This is the standard.”</i>	baseline framework
NON-NORMATIVE	<i>“This reflects the standard.”</i>	reflected interpretation
CRITICAL	<i>“This inverts the standard.”</i>	inverted assumptions
METACOGNITIVE	<i>“The observer audits the frame.”</i>	observer audits its own evaluative frame — thinking about thinking

Note on the METACOGNITIVE phrasing. The earlier line *“This audits the standard”* is deprecated as of v1.3.0-rc. The audit at METACOGNITIVE is directed at the **observer’s**



own evaluative frame, not at the standard as an external object. The explanatory variant — “*Thinking examines its own frame*” — appears in the warmer-voiced OWL-SEMAPHORE-EXPLANATION.md. This refinement aligns the language with the cognitive-science meaning of metacognition: monitoring and regulation of one’s own cognitive process (metacognitive reflection review, PMC 11368986).

4.3 Critical Distinction

The system is only valid if these states are not conflated.

5. Physical Grounding

The system is grounded in observable behavior.

5.1 Canonical Example

The METACOGNITIVE state is physically instantiated by:

- searching a space normally
- failing to detect a target
- inverting the viewing frame

5.2 Interpretation

The object does not change. The observer does not change as an agent. The frame changes.

In short: the observer audits the frame.

6. Visual System Mapping

6.1 Shared Geometry

All owls share:

- identical center
- identical radial structure
- identical meander ring

6.2 Transform Consistency

Each owl is derived from the normative form by a valid element of V_4 .

No arbitrary transformations are permitted.



6.3 Invariants

- geometry is fixed
 - center is fixed
 - ring structure is fixed
-

7. Color System and Accessibility

Each state is assigned a distinct color space region:

- gold → normative authority
- teal → analytical reflection
- red → adversarial inversion
- amethyst → introspective frame-audit

7.1 Constraint

Color is semantic, not decorative.

7.2 Accessibility — Triple-Redundant Encoding (v1.3.0-rc, normative)

Color is not the only carrier. State identity must remain recoverable when color is removed (grayscale rendering, color vision deficiency, or low-vision contexts). Every state in the Owl Semaphore is therefore encoded through at least three independent channels:

1. **color** (the palette assigned in §7)
2. **orientation** (the V_4 transform applied to the canonical owl: upright/inverted × right/left-facing)
3. **textual label and context** (the literal state token — **NORMATIVE**, **NON-NORMATIVE**, **CRITICAL**, **METACOGNITIVE** — and the math/quote tuple printed alongside the badge)

This satisfies the design intent of **WCAG 2.2 SC 1.4.1 (Use of Color, Level A)**, which prohibits color from being the only visual means of conveying information (W3C), and aligns with Section 508 §302.3 (Section 508.gov). The **CRITICAL** state’s intentionally low red-on-red contrast is the most acute test of this rule: redness alone never carries **CRITICAL** identity — orientation (upside-down, left-facing) and the literal label **CRITICAL** are required.

Red-green color vision deficiency affects approximately 8% of males and 0.5% of females of Northern European descent; rates vary by population (PMC global CVD review, 12385717).

Full WCAG 2.2 Level AA empirical conformance testing (automated checks, CVD simulation, user testing) is scoped to a future release; v1.3.0-rc states the design rule and its compliance intent.

8. Integrity Model



All assets must satisfy:

- reproducibility from layers
- RGBA transparency correctness
- cryptographic verification (SHA-3-512)

8.1 Principle

An asset is not valid because it looks correct. It is valid because it verifies.

9. File and System Architecture

9.1 Structure

OWL-SEMAPHORE/ |— OWL-SEMAPHORE-SYSTEM.md |— OWL-SEMAPHORE-EXPLANATION.md |— INTEGRITY-MANIFEST.md |— CHANGELOG.md |— OWL-1-NORMATIVE.md |— OWL-2-NON-NORMATIVE.md |— OWL-3-CRITICAL.md |— OWL-4-METACOGNITIVE.md

9.2 Separation of Concerns

- system rules live in the system spec
 - state rules live in the four owl-specific files
 - origin story and audience rationale live in the explanation document
 - canonical sentence history lives in `CHANGELOG.md`
-

10. Interpretation Doctrine

10.1 The System Encodes Position, Not Truth

The Owl Semaphore does not assert correctness.

It encodes:

- how something is being evaluated
- not whether it is ultimately true

10.2 Misuse Condition

If the system is used to imply certainty rather than epistemic position, it is being used incorrectly.

11. Core Principle (Reconciled, v1.3.0-rc)



This system is defined as:

A finite algebra over epistemic states, implemented as a reproducible visual notation system with enforced invariants.

This sentence is the single formal canonical definition for v1.3.0-rc. It supersedes both “*implemented as a reproducible visual system with enforced invariants*” and “*mapped into a visual system with strict invariants*”. See CHANGELOG.md for the per-version canonical-sentence history.

12. Normative-Language Discipline

Where the Owl Semaphore documents use the RFC 2119 / RFC 8174 keywords (MUST, MUST NOT, SHALL, SHALL NOT, SHOULD, SHOULD NOT, RECOMMENDED, NOT RECOMMENDED, MAY, OPTIONAL), they carry their BCP 14 meaning **only when they appear in all capitals** (RFC 2119; RFC 8174). Lowercase forms (“must”, “should”, “may”) carry ordinary English meanings.

Sections labeled “(normative)” form part of the canonical specification; sections labeled “(informative)” or “(explanatory)” provide context only and do not extend the algebra.

13. Closing Statement

The Owl Semaphore is designed to make reasoning visible.

It enforces structure where ambiguity would otherwise dominate, and it preserves interpretability across transformation, disagreement, and self-examination.

It is not decoration.

It is a constraint system for thought.



OWL SEMAPHORE SYSTEM — CLASSIFICATION LEDGER (v1.3.0-rc)



NORMATIVE

$T = I \quad \det = +1$

$(x, y) \rightarrow (x, y)$

“This is the standard.”

RFC 2119 MUST / SHALL



NON-NORMATIVE

$T = \sigma_v \quad \det = -1$

$(x, y) \rightarrow (-x, y)$

“This reflects the standard.”

Informative / Advisory (NOTE)



CRITICAL

$T = C_2 \quad \det = +1$

$(x, y) \rightarrow (-x, -y)$

“This inverts the standard.”

RFC 2119 MUST NOT / SHALL NOT



METACOGNITIVE

$T = \sigma_h \quad \det = -1$

$(x, y) \rightarrow (x, -y)$

“The observer audits the frame.”

Epistemic / Framework (META)

Accessibility: color is not the only carrier. State identity is recoverable from color + orientation + textual label (WCAG 2.2 SC 1.4.1).

Owl Semaphore v1.3.0-rc · github.com/IT-Help-San-Diego/owl-semaphore
Concept DOI 10.5281/zenodo.19473697 · v1.2.0 DOI 10.5281/zenodo.19474599 · v1.3.0 DOI TBD_BY_ZENODO_ON_RELEASE
(c) 2024-2026 IT Help San Diego Inc. · Licensed under CC-BY-4.0